Supplementary Material

Quantum Machine Learning for Cyber-Physical Anomaly Detection in Unmanned Aerial Vehicles: Lessons from a Proxy-Audited Temporal Evaluation

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This document holds the detailed tables behind the article. Unless noted, values are means with standard deviations in parentheses over the ten partitions ("seeds") of the same log. The standard deviation measures sensitivity to the choice of partition. It is not a confidence interval for performance on an independent flight. *all*, *reduced*, and *physical* denote the three input sets (72, 63, and 22 variables). The false-alarm rate is the fraction of nominal samples flagged as faults; the miss rate is the fraction of fault samples labeled nominal. All models use default decision thresholds.

S1 Data, labels, and partitions

The table rebuilt from the raw sensor files has 4 817 rows (about 10 Hz) and 81 numeric variables, of which nine are constant and were dropped. The timestamps run on one clock for 506 s. Two logging gaps of 16.3 s and 8.4 s separate three segments of 144, 4 606, and 67 rows; the first segment is entirely nominal and the last entirely remote-control fault. There are four contiguous fault intervals: label 1 from 92.0 to 152.7 s (608 rows), label 2 from 226.0 to 285.3 s (594 rows), label 3 from 335.0 to 383.1 s (482 rows), and label 4 from 430.9 to 506.1 s (670 rows).

Check of the label meanings. The data set's authors describe four simulated failures in the order GPS, accelerometer, engine, and remote control. The raw signals agree with labels 1 to 4 in that order. During label 1 the reported satellite count drops from 10 to 3. During label 2 all three accelerometer channels are constant. During label 3 the ground speed is zero, the altitude estimate is at ground level, and the throttle output is saturated. During label 4 the vehicle descends.

Table S1 gives the ten time blocks and Table S2 the ten partitions. In every partition some training rows are later in time than the earliest test row, and at least one test block borders a training block. Blocks 0 and 7 are never used for testing; block 8 is used in six partitions. In seven partitions at least one fault interval in the test blocks contributes no rows to training. The median absolute autocorrelation of the 22 physical variables is 0.87 at 0.1 s, 0.67 at 1 s, 0.46 at 10 s, and 0.07 at 48 s (one block).

Table S1: The ten time blocks and their label counts.

Block	Rows	Start (s)	End (s)	Nominal	GPS	Accel.	Engine	RC
0	482	0.0	64.3	482	0	0	0	0
1	482	64.4	112.5	276	206	0	0	0
2	482	112.6	160.7	80	402	0	0	0
3	481	160.8	208.8	481	0	0	0	0
4	482	208.9	257.0	171	0	311	0	0
5	482	257.1	305.2	199	0	283	0	0
6	481	305.3	353.3	297	0	0	184	0
7	482	353.4	401.5	184	0	0	298	0
8	482	401.6	449.7	293	0	0	0	189
9	481	449.8	506.1	0	0	0	0	481

S2 Complete results of the blocked evaluation

Tables S3–S6 give all twelve models and all three input sets. In the stored result files the column `far_class0` is the miss rate; the false-alarm rate was recovered from the same files as $2\times$`far_macro`$-$`far_class0` and checked against recomputed confusion matrices for the classical models (maximum difference 2×10^{-16}).

Table S2: The ten partitions. Validation blocks are the two blocks not listed.

Seed	Training blocks	Test blocks	Fault share in test	Training rows later than first test row	Test-block edges touching training	Fault types in test	Test fault rows from intervals unseen in training
0	0, 1, 4, 5, 6, 9	2, 8	0.61	67%	2	1, 4	0
1	0, 1, 4, 5, 6, 7	2, 9	0.92	67%	1	1, 4	481
2	0, 3, 5, 7, 8, 9	1, 4	0.54	83%	3	1, 2	206
3	0, 1, 2, 3, 6, 8	4, 5	0.62	33%	2	2	594
4	0, 1, 2, 4, 6, 7	3, 8	0.20	50%	3	4	189
5	0, 1, 3, 4, 6, 7	5, 9	0.79	33%	2	2, 4	481
6	2, 3, 4, 5, 7, 9	1, 8	0.41	100%	3	1, 4	0
7	0, 1, 3, 4, 7, 9	5, 8	0.49	33%	3	2, 4	0
8	1, 2, 3, 4, 5, 7	6, 8	0.39	17%	3	3, 4	189
9	0, 2, 3, 5, 6, 7	4, 8	0.52	50%	3	2, 4	189

Table S3: F1 score (macro average).

Model	all	reduced	physical
Logistic regression	0.65 (0.17)	0.65 (0.17)	0.51 (0.14)
SVM	0.58 (0.18)	0.58 (0.18)	0.46 (0.20)
MLP	0.64 (0.16)	0.65 (0.17)	0.53 (0.14)
Random forest	0.56 (0.13)	0.57 (0.14)	0.54 (0.15)
XGBoost	0.52 (0.14)	0.54 (0.14)	0.52 (0.13)
DRU circuit alone	0.57 (0.22)	0.51 (0.16)	0.51 (0.20)
XGB only (5 variables)	0.35 (0.10)	0.47 (0.22)	0.52 (0.15)
XGB + principal components	0.37 (0.07)	0.49 (0.23)	0.53 (0.19)
XGB + polynomial terms	0.40 (0.13)	0.50 (0.16)	0.53 (0.15)
XGB + random RBF features	0.50 (0.18)	0.44 (0.15)	0.51 (0.14)
XGB + untrained DRU output	0.34 (0.14)	0.47 (0.20)	0.54 (0.16)
XGB + trained DRU output	0.51 (0.22)	0.53 (0.19)	0.56 (0.14)

Table S7 compares the trained-circuit hybrid with every other model partition by partition (physical inputs). Each entry counts the partitions in which the hybrid is better / tied / worse.

Table S4: Area under the ROC curve.

Model	all	reduced	physical
Logistic regression	0.66 (0.30)	0.66 (0.30)	0.55 (0.15)
SVM	0.67 (0.20)	0.68 (0.19)	0.58 (0.19)
MLP	0.69 (0.29)	0.68 (0.28)	0.62 (0.30)
Random forest	0.79 (0.18)	0.77 (0.15)	0.73 (0.11)
XGBoost	0.67 (0.16)	0.70 (0.17)	0.67 (0.15)
DRU circuit alone	0.76 (0.26)	0.57 (0.31)	0.56 (0.27)
XGB only (5 variables)	0.45 (0.19)	0.49 (0.21)	0.56 (0.22)
XGB + principal components	0.62 (0.21)	0.66 (0.22)	0.60 (0.23)
XGB + polynomial terms	0.63 (0.20)	0.52 (0.17)	0.67 (0.16)
XGB + random RBF features	0.80 (0.16)	0.69 (0.15)	0.56 (0.22)
XGB + untrained DRU output	0.53 (0.22)	0.49 (0.21)	0.57 (0.23)
XGB + trained DRU output	0.75 (0.22)	0.54 (0.27)	0.61 (0.17)

Table S5: False-alarm rate.

Model	all	reduced	physical
Logistic regression	0.16 (0.16)	0.16 (0.16)	0.31 (0.30)
SVM	0.16 (0.18)	0.16 (0.16)	0.32 (0.30)
MLP	0.21 (0.17)	0.17 (0.17)	0.26 (0.28)
Random forest	0.26 (0.31)	0.27 (0.29)	0.27 (0.29)
XGBoost	0.39 (0.33)	0.38 (0.31)	0.28 (0.30)
DRU circuit alone	0.33 (0.32)	0.36 (0.40)	0.42 (0.28)
XGB only (5 variables)	0.50 (0.53)	0.29 (0.34)	0.26 (0.30)
XGB + principal components	0.37 (0.48)	0.25 (0.35)	0.36 (0.30)
XGB + polynomial terms	0.44 (0.49)	0.43 (0.41)	0.28 (0.29)
XGB + random RBF features	0.20 (0.35)	0.25 (0.33)	0.34 (0.29)
XGB + untrained DRU output	0.40 (0.52)	0.34 (0.40)	0.29 (0.32)
XGB + trained DRU output	0.34 (0.39)	0.32 (0.34)	0.34 (0.30)

Table S6: Miss rate.

Model	all	reduced	physical
Logistic regression	0.43 (0.29)	0.44 (0.30)	0.52 (0.32)
SVM	0.52 (0.33)	0.52 (0.33)	0.55 (0.41)
MLP	0.40 (0.34)	0.42 (0.33)	0.51 (0.36)
Random forest	0.45 (0.36)	0.45 (0.36)	0.48 (0.36)
XGBoost	0.46 (0.36)	0.44 (0.36)	0.51 (0.34)
DRU circuit alone	0.38 (0.36)	0.53 (0.25)	0.51 (0.19)
XGB only (5 variables)	0.60 (0.44)	0.54 (0.40)	0.53 (0.32)
XGB + principal components	0.62 (0.45)	0.53 (0.42)	0.46 (0.39)
XGB + polynomial terms	0.54 (0.42)	0.45 (0.35)	0.52 (0.33)
XGB + random RBF features	0.57 (0.39)	0.63 (0.37)	0.51 (0.35)
XGB + untrained DRU output	0.62 (0.45)	0.51 (0.37)	0.49 (0.35)
XGB + trained DRU output	0.50 (0.34)	0.55 (0.29)	0.45 (0.27)

Table S7: Trained-circuit hybrid against each other model: partitions won / tied / lost (physical inputs).

Compared with	F1	ROC AUC	False-alarm rate	Miss rate
DRU circuit alone	6/0/4	5/0/5	5/0/5	7/0/3
Logistic regression	6/0/4	7/0/3	5/0/5	4/1/5
MLP	4/0/6	4/0/6	5/0/5	4/1/5
Random forest	4/0/6	2/0/8	3/1/6	3/2/5
SVM	7/0/3	8/0/2	6/0/4	4/1/5
XGB + untrained DRU output	5/0/5	5/0/5	2/2/6	5/1/4
XGB + principal components	4/0/6	3/0/7	5/1/4	4/1/5
XGB + polynomial terms	5/0/5	4/0/6	2/1/7	7/1/2
XGB + random RBF features	7/0/3	6/0/4	6/1/3	4/2/4
XGB only (5 variables)	5/0/5	4/0/6	2/2/6	5/1/4
XGBoost	7/0/3	4/0/6	4/1/5	5/1/4

S3 Effect of the split design

Table S8 repeats the classical experiments with a random stratified split of rows (70/15/15, ten seeds) and with the blocked split after deleting every training row within a given time gap of any test row. The average number of real training rows falls from 2 890 (no gap) to 2 765, 2 515, 2 152, and 1 734, so the gap analysis confounds reduced adjacency with reduced training data.

Table S8: Mean F1 of the classical models under different splits.

Inputs	Model	Random rows	Blocks	5 s gap	15 s gap	30 s gap	48 s gap
all	Logistic regression	0.99	0.65	0.58	0.58	0.34	0.46
all	SVM	0.95	0.58	0.50	0.51	0.39	0.37
all	MLP	0.99	0.64	0.54	0.55	0.36	0.38
all	Random forest	1.00	0.56	0.51	0.43	0.28	0.36
all	XGBoost	1.00	0.52	0.48	0.39	0.30	0.35
all	Mean of five	0.98	0.59	0.52	0.49	0.33	0.38
physical	Logistic regression	0.83	0.51	0.44	0.40	0.27	0.31
physical	SVM	0.77	0.46	0.44	0.42	0.34	0.32
physical	MLP	0.97	0.53	0.46	0.43	0.36	0.33
physical	Random forest	0.99	0.54	0.51	0.45	0.34	0.34
physical	XGBoost	0.99	0.52	0.47	0.40	0.34	0.35
physical	Mean of five	0.91	0.51	0.46	0.42	0.33	0.33

Training on the past only. Models were trained on blocks $0, \ldots, k-1$ and tested on block k for $k = 2, \ldots, 9$. Table S9 gives the accuracy per test block with physical inputs and Table S10 the pooled error rates. On block 4, the first block of the accelerometer fault, no model detects a single fault sample.

Table S9: Past-only training: accuracy on each test block (physical inputs).

Test block	Fault share	Logistic regression	SVM	MLP	Random forest	XGBoost
2	0.83	0.71	0.62	0.82	0.83	0.83
3	0.00	0.64	0.67	0.57	1.00	0.64
4	0.65	0.35	0.35	0.35	0.35	0.35
5	0.59	0.45	0.91	0.83	0.92	0.50
6	0.38	0.92	0.93	0.81	0.80	0.84
7	0.62	0.95	0.45	0.96	0.79	0.68
8	0.39	0.61	0.61	0.61	0.61	0.48
9	1.00	0.05	0.06	0.05	0.06	0.04

Table S10: Past-only training: results pooled over test blocks 2–9.

Inputs	Model	F1	False-alarm rate	Miss rate
all	Logistic regression	0.70	0.26	0.34
all	MLP	0.60	0.23	0.53
all	Random forest	0.70	0.27	0.32
all	SVM	0.58	0.39	0.45
all	XGBoost	0.59	0.27	0.52
physical	Logistic regression	0.59	0.27	0.52
physical	MLP	0.62	0.17	0.54
physical	Random forest	0.67	0.13	0.49
physical	SVM	0.56	0.12	0.66
physical	XGBoost	0.54	0.27	0.60

Holding out gap-delimited segments is not informative: the two short segments contain a single class, and the long one leaves 211 training rows.

S4 Detection by exposure to the fault event

For each partition, test fault samples were divided according to whether their fault interval also contributed rows to training. Table S11 pools the ten partitions (14 745 and 11 645 sample-level decisions per input set

across the five models).

Table S11: Fraction of fault samples detected.

Inputs	Model	Interval partly in training	Interval absent from training
all	Logistic regression	0.92	0.18
all	SVM	0.79	0.08
all	MLP	0.94	0.26
all	Random forest	0.97	0.03
all	XGBoost	0.96	0.03
physical	Logistic regression	0.83	0.04
physical	SVM	0.65	0.12
physical	MLP	0.88	0.01
physical	Random forest	0.93	0.03
physical	XGBoost	0.89	0.03

S5 What each model received, and controls

Classical models: all variables of the input set; all training rows after rebalancing (about 2 900–3 800, of which 2 890 are real); default settings; fitted in under two seconds. *Circuit:* the five variables ranked highest by mutual information on the training blocks; subset A, 800 rows (400 per class); 30 weights; 200 optimizer iterations. *Hybrids and controls:* the same five variables plus 0 to 20 derived columns; subset B, 800 rows disjoint from A; XGBoost with 100 trees. *Trained-circuit hybrid:* its extra column comes from a circuit fitted on the labels of subset A, so it draws on 1 600 labeled rows, against 800 for the controls. The validation blocks were not used. Subsets A and B are drawn from the rebalanced training set, in which 12% of rows on average are synthetic, so they are disjoint but not independent.

"Identical budget" in the original submission meant the same partitions and test blocks for all models, and the same subset B and XGBoost settings for the six hybrids. It did not mean equal samples, inputs, computation, or capacity across model families.

Table S12 traces the difference between classical XGBoost and the "XGB only" control step by step, and adds two label-informed classical controls. With all variables, the five selected variables are accumulating quantities whose test values fall outside the training range in 30% of test rows on average; the "XGB only" control then predicts a single class in seven of ten partitions.

Table S12: Mean F1 (standard deviation) for the fairness decomposition and added controls.

Configuration	all	reduced	physical
Classical XGBoost: all variables of the set, all training rows	0.52 (0.14)	0.54 (0.14)	0.52 (0.13)
Same, restricted to the five selected variables	0.42 (0.16)	0.43 (0.22)	0.55 (0.13)
Same, restricted to the 800-row subset B	0.35 (0.10)	0.47 (0.22)	0.53 (0.15)
Same, with the hybrid learning rate (this is the "XGB only" control)	0.35 (0.10)	0.47 (0.22)	0.52 (0.15)
XGB + output of logistic regression fitted on subset A	0.36 (0.13)	0.46 (0.21)	0.48 (0.17)
XGB + output of a 29-weight MLP fitted on subset A	0.34 (0.08)	0.44 (0.20)	0.51 (0.17)
XGB + trained DRU output (stored result)	0.51 (0.22)	0.53 (0.19)	0.56 (0.14)

S6 Sensitivity to synthetic oversampling

SMOTETomek was applied after the split, to training rows only. Table S13 repeats the classical experiments without it. Table S14 repeats the circuit and hybrid experiments on real rows only (physical inputs; ten new circuit fits). Table S15 characterizes the synthetic rows. On average 9% of them (up to 23%) have a magnetic-field magnitude outside the central 98% range of the real training rows, where 2% would be expected. We make no claim that the synthetic samples are physically plausible.

Table S13: Classical models with and without SMOTETomek (means over ten partitions).

Model	Inputs	F1 with	F1 without	AUC with	AUC without
Logistic regression	all	0.650	0.643	0.655	0.650
Logistic regression	reduced	0.649	0.649	0.660	0.660
Logistic regression	physical	0.506	0.508	0.551	0.542
MLP	all	0.641	0.638	0.693	0.671
MLP	reduced	0.651	0.659	0.679	0.680
MLP	physical	0.528	0.531	0.619	0.617
Random forest	all	0.564	0.557	0.786	0.775
Random forest	reduced	0.566	0.564	0.773	0.781
Random forest	physical	0.545	0.538	0.731	0.715
SVM	all	0.577	0.575	0.670	0.667
SVM	reduced	0.584	0.583	0.683	0.680
SVM	physical	0.462	0.469	0.581	0.581
XGBoost	all	0.521	0.504	0.669	0.657
XGBoost	reduced	0.538	0.522	0.704	0.706
XGBoost	physical	0.524	0.516	0.669	0.648

Table S14: Circuit and hybrids without synthetic rows (physical inputs).

Model	F1 with resampling	F1 without	ROC AUC without	False-alarm rate without	Miss rate without
DRU circuit alone	0.51 (0.20)	0.50 (0.18)	0.51 (0.27)	0.46 (0.29)	0.50 (0.25)
XGB only (5 variables)	0.52 (0.15)	0.52 (0.13)	0.57 (0.22)	0.26 (0.30)	0.54 (0.34)
XGB + principal components	0.53 (0.19)	0.47 (0.18)	0.62 (0.20)	0.29 (0.33)	0.59 (0.36)
XGB + polynomial terms	0.53 (0.15)	0.52 (0.16)	0.71 (0.19)	0.28 (0.30)	0.53 (0.33)
XGB + random RBF features	0.51 (0.14)	0.52 (0.18)	0.54 (0.26)	0.34 (0.34)	0.48 (0.37)
XGB + untrained DRU output	0.54 (0.16)	0.49 (0.15)	0.55 (0.24)	0.32 (0.32)	0.57 (0.31)
XGB + trained DRU output	0.56 (0.14)	0.56 (0.14)	0.66 (0.19)	0.33 (0.38)	0.45 (0.35)
XGB + logistic-regression output (control)	–	0.49 (0.15)	0.56 (0.25)	0.28 (0.34)	0.59 (0.31)
XGB + 29-weight MLP output (control)	–	0.50 (0.14)	0.59 (0.24)	0.37 (0.38)	0.53 (0.29)

Table S15: Synthetic rows in the rebalanced training sets (physical inputs).

Seed	Real training rows	Synthetic rows	Synthetic share	Magnetic-field norm outside real 1–99% range	Acceleration norm outside real 1–99% range
0.0	2890.0	40.0	0.01	0.00	0.03
1.0	2891.0	326.0	0.10	0.08	0.03
2.0	2890.0	388.0	0.12	0.14	0.01
3.0	2890.0	928.0	0.24	0.06	0.04
4.0	2891.0	89.0	0.03	0.04	0.01
5.0	2890.0	892.0	0.24	0.05	0.02
6.0	2890.0	660.0	0.19	0.23	0.01
7.0	2890.0	297.0	0.09	0.10	0.01
8.0	2891.0	109.0	0.04	0.12	0.03
9.0	2890.0	556.0	0.16	0.08	0.01

S7 Cost of the simulated circuit

Table S16 reports measurements on the classical computer that ran the simulation. They do not describe a quantum processor. On hardware, each expectation value would be estimated from repeated circuit runs affected by gate and readout noise, with queueing and communication delays; none of these was measured. The 30 circuit fits of the original study took 117 minutes in total. A second measurement of the same fit gave 234 s and 1.4 ms per sample; all other values were identical.

Table S16: Measured cost of one circuit fit and of inference in simulation.

Quantity	Value
Qubits / trainable weights / circuit depth	5 / 30 / 16
Gates	10 rx, 10 ry, 10 rz, 10 cx
Simulator	exact state vector (Qiskit StatevectorEstimator), no shots, no noise
Host	Apple M5, 10 cores, 16 GB RAM; macOS-27.0.1-arm64-arm-64bit
Training rows per fit	800
Optimizer iterations (COBYLA)	200
Loss evaluations per fit (measured)	328
Simulated circuit runs per fit (measured)	262 400
Wall-clock time per fit (measured; stored run)	227.8 s; 229.9 s
Training loss, first / last evaluation	1.0072 / 0.6327
Inference, one sample, circuit (median / 95th percentile)	1.28 / 1.48 ms
Inference, one sample, hybrid (median / 95th percentile)	1.62 / 1.86 ms
Inference, one sample, XGBoost alone (median)	0.25 ms
Stored result reproduced (F1, seed 0, physical-only)	0.1362 vs. 0.1362

S8 Secondary experiment on the engine fault

A one-fault task (nominal versus engine fault) was run with twenty time blocks and XGBoost. In nine of the ten random assignments at least one of the training, validation, and test sets contained no fault sample, and those assignments were skipped; one remained (seed 1). On that single partition F1 was 0.93 with the *all* and *reduced* inputs and 0.43 with *physical* inputs, where every test row was labeled nominal although the ROC AUC stayed at 0.96. A single partition supports no estimate and is reported only as an anecdote.

S9 Reproducibility

Every stored result that was rerun was reproduced exactly: all 150 classical runs, the 30 runs of the five-variable XGBoost control, and one circuit fit (seed 0, *physical* inputs; Table S16). The other 29 circuit fits of the main study, and the hybrid variants that depend on them, were not rerun. The circuit and hybrid rows of Tables S3–S6 and the comparisons in Table S7 are therefore computed from the stored per-partition files of the original study; the false-alarm rate in those rows is derived arithmetically from two stored columns. All other tables come from analyses executed for this revision. Scripts, per-partition outputs, software versions, and commands are in the public repository `https://github.com/Carlosandp/TLM-UAV-Quantum-Anomaly-Detection` (folder `revision/`).